

Digital Delta

Offline relief coordination for Bangladesh

Innovate to Market

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Innovation Fair 2026



Concept illustration

Flood outages disrupt the delivery record

Transport, power and communications can fail together.

7.2 million

People affected
June 2022, nine districts

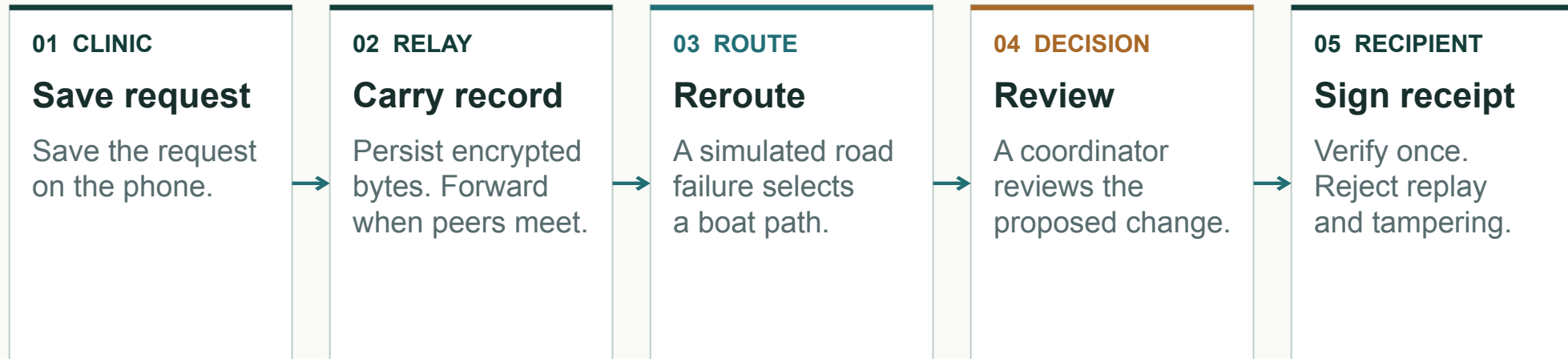
3.74 million

People affected
June 2024, six districts

A request still needs to reach the next responsible person when the network is unavailable.

One delivery journey connects the modules

Clinic request, relay, rerouting, human review and signed receipt.

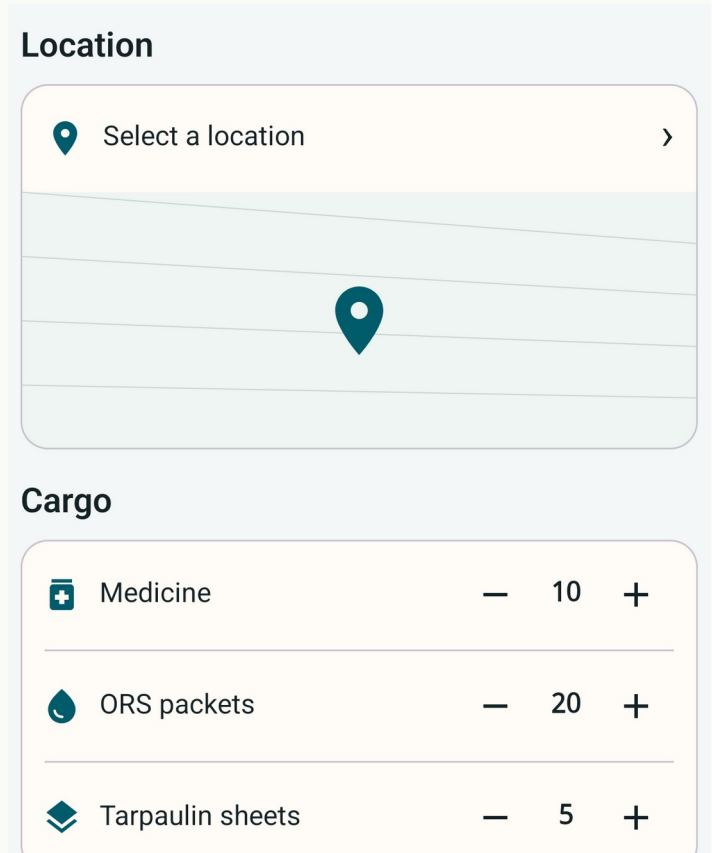
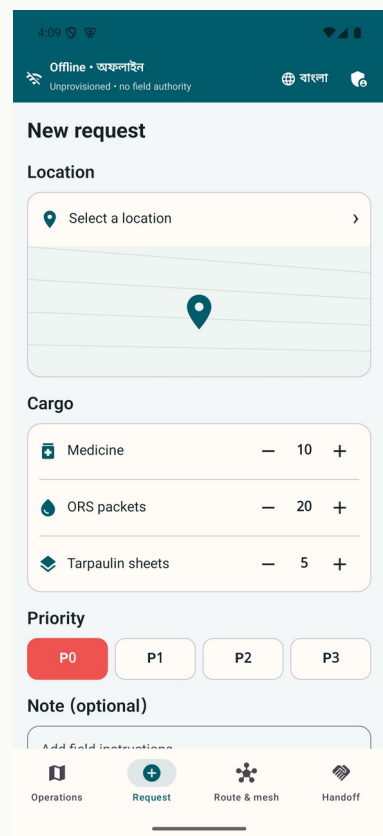


A conceptual walkthrough across demonstrated modules

Screenshots come from separate seeded captures. Physical end-to-end acceptance remains open.

The clinic can record a request offline

The phone owns the local record before any transfer.



01 Structured request

Location, supplies and urgency belong to a record the phone can retain.

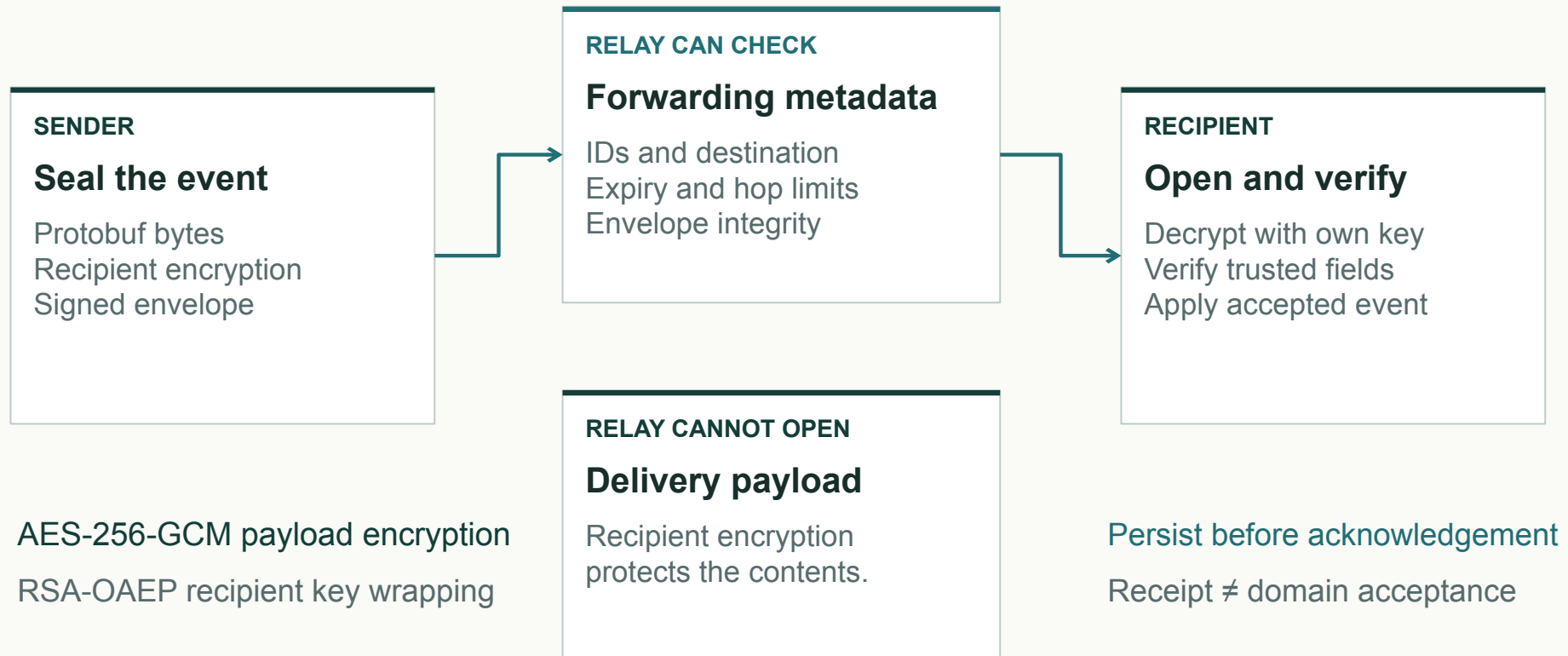
02 Honest transfer state

Queued, forwarded and accepted remain separate states.

Left: full source capture. Center: enlarged detail crop. Both show the actual application.

The relay forwards encrypted contents

Only the intended recipient opens the protected delivery payload.



Metadata remains available for forwarding. Payload encryption does not hide every communication pattern.

A failed road changes the phone's route

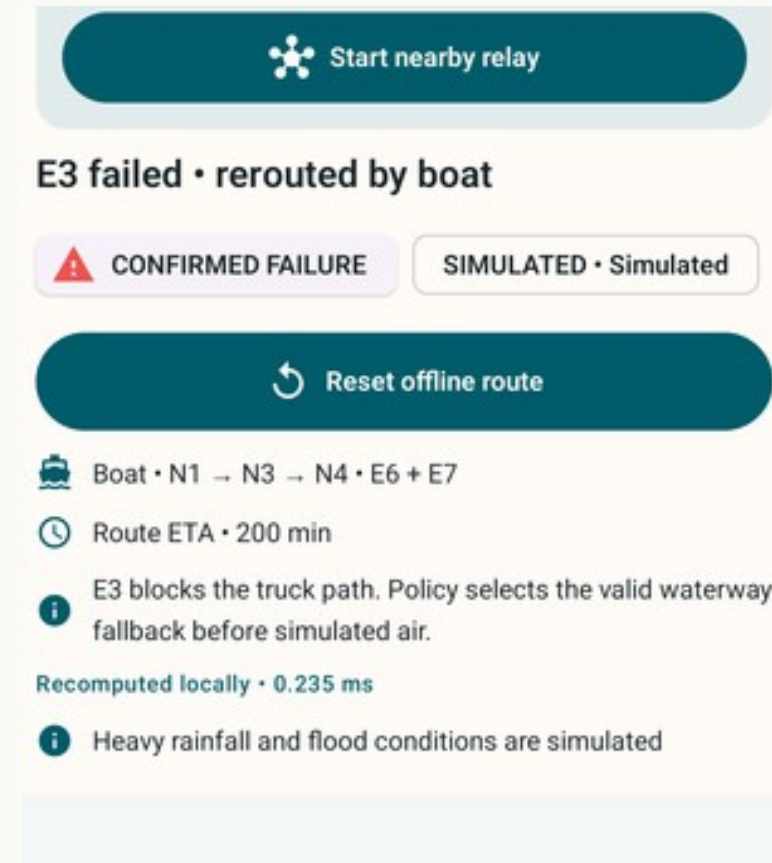
The seeded E3 failure selects the boat alternative E6 and E7.

Truck

E1 and E3

Boat after failure

E6 and E7



Enlarged source detail. Route failure and vehicles are simulated. Target-phone performance requires measurement.

A coordinator confirms the priority change

The proposal explains the deadline pressure and safe drop.

Recomputed locally • 0.249 ms

**P0 SLA breach predicted**
30% • 295 / 120 min

SIMULATED • Simulated

Baseline arrival • 235 min

Deposit P2 tarpaulin at Sunamganj Sadar Camp

Estimated time gained • 25 min

Assignment change requires coordinator confirmation.

 **Confirm P2 drop and prioritize P0**

 Heavy rainfall and flood conditions are simulated

01 Explain the tradeoff

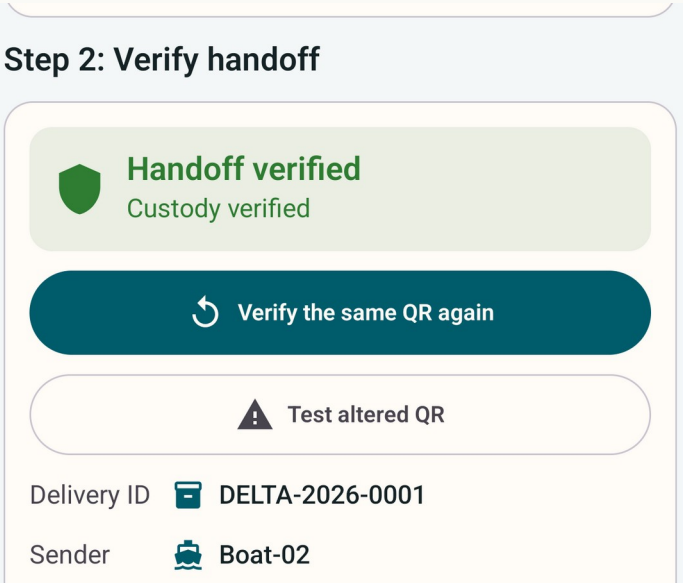
The longer boat route threatens the urgent delivery deadline.

02 Retain the decision

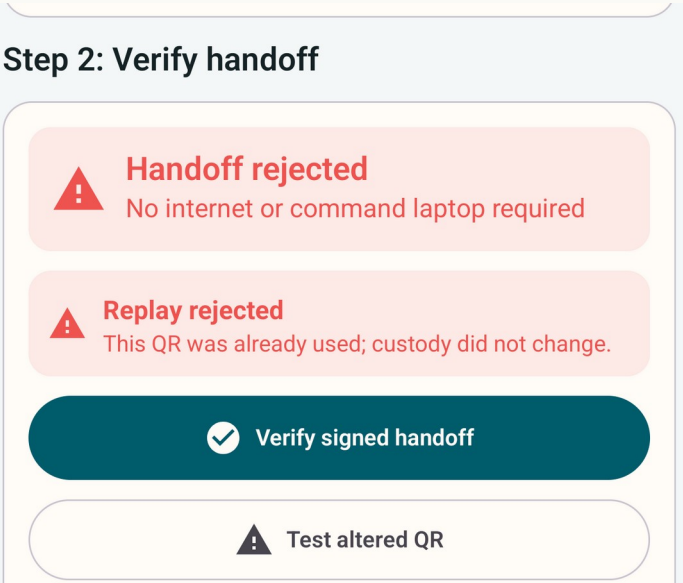
A person confirms the assignment change and its reason.

A signed handoff accepts custody once

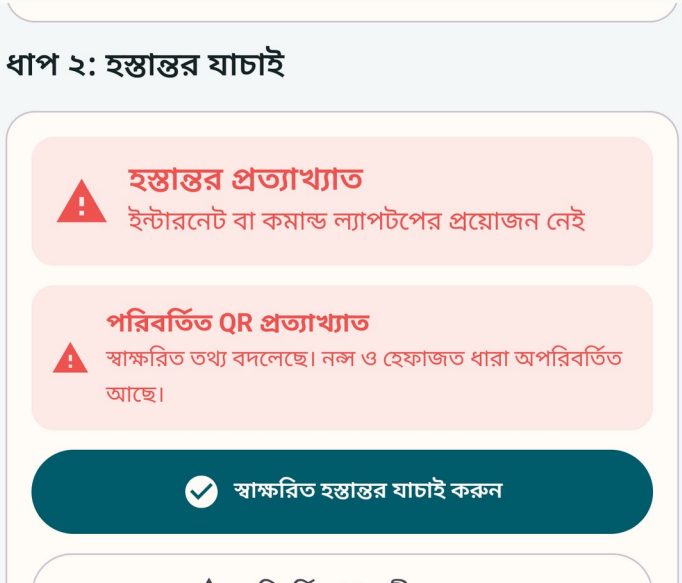
Replay and altered offers must leave custody unchanged.



Accepted



Replay rejected



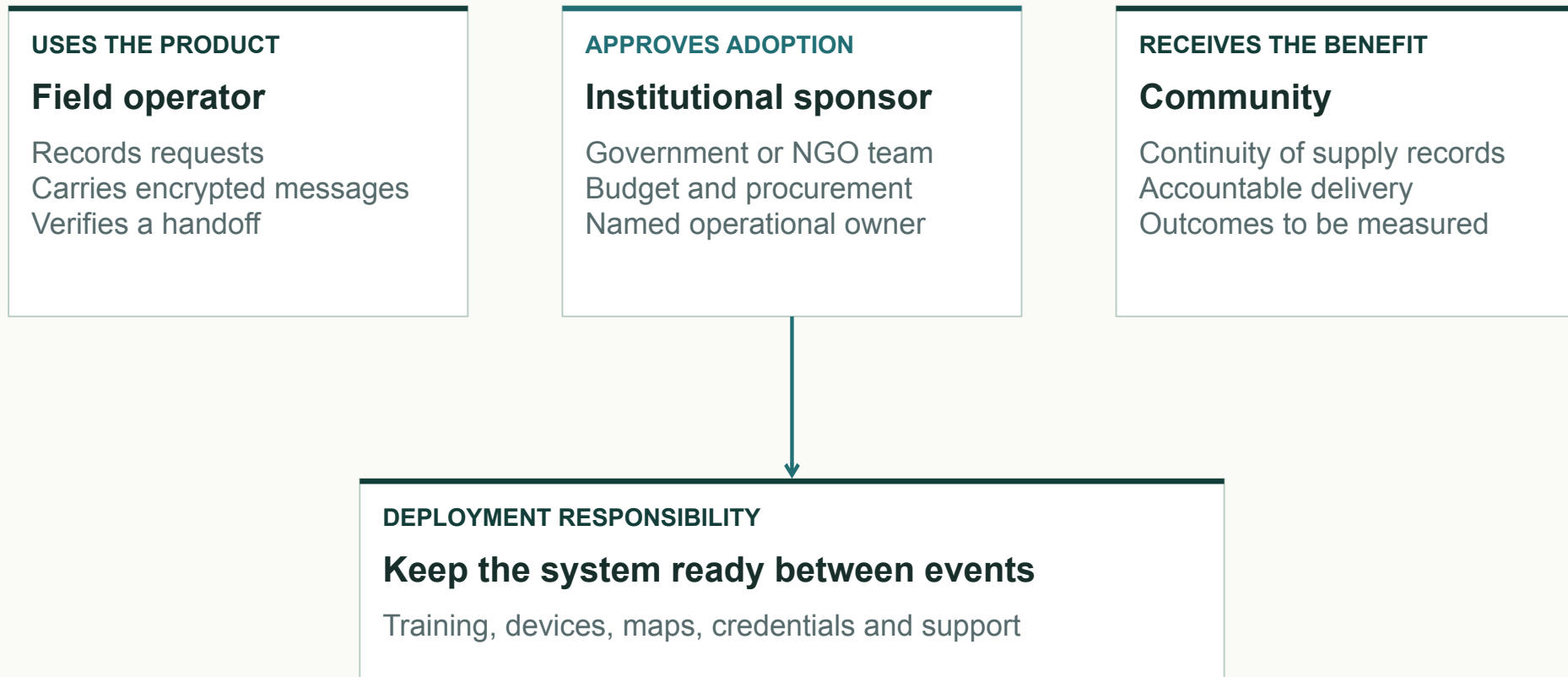
Tamper rejected

Enlarged source details. Full captures appear in the report. Physical camera acceptance remains open.

The first buyer owns preparedness

An institutional sponsor needs a named operational owner.

Buyer hypothesis. Interviews and willingness to pay remain unvalidated.



Proposed roles. Field users, budget owners and community beneficiaries are different groups.

The integrated workflow must earn adoption

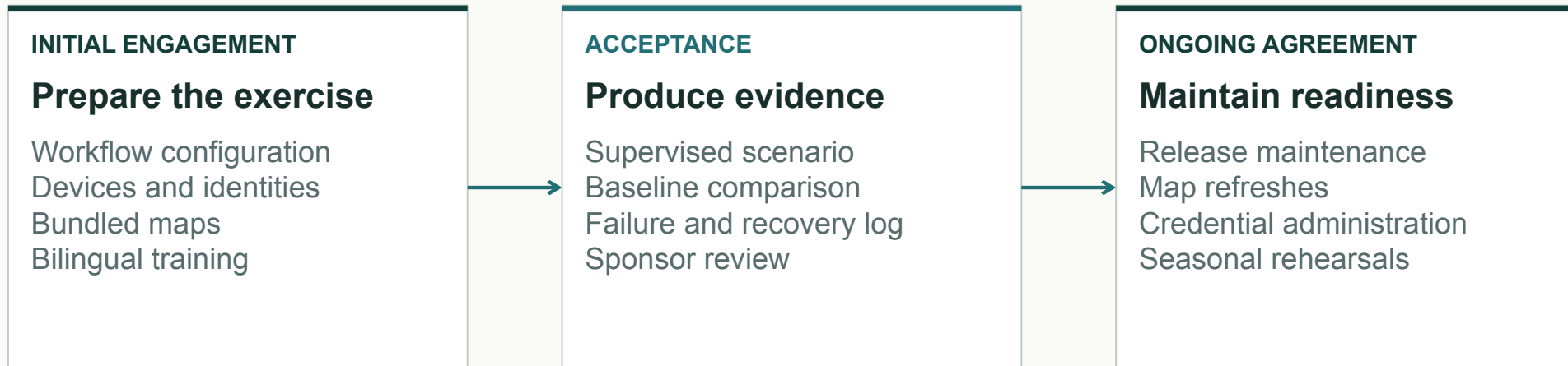
The comparison starts with each alternative's documented strength.

Alternative	Documented strength	Digital Delta adoption question
Sahana Eden	Relief inventory and logistics	Does phone continuity add value?
KoboCollect	Offline assessment collection	Does custody need a separate workflow?
Ushahidi	Reports and incident mapping	Can verified events complement reporting?
Briar	Encrypted nearby messaging	Are structured logistics rules worth training?
Meshtastic	LoRa mesh messaging	Can existing phones meet the relay need?

Positioning hypothesis. A missing feature in reviewed documentation is not proof that a competitor cannot provide it.

The offer is a preparedness service

Configuration, measured exercises and agreed support surround the software.



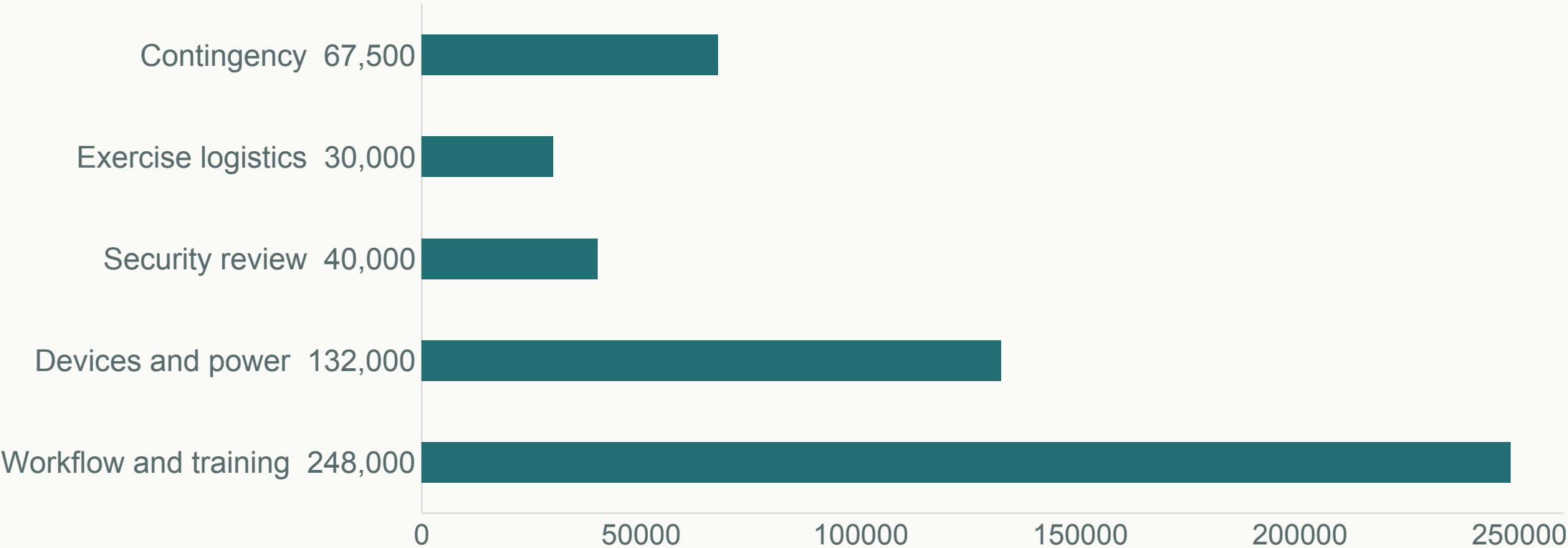
The proposed commercial unit is an institutional service engagement.

Pricing and repeat demand require validation. Sensitive data is not a revenue product.

Initial engagement and recurring support are commercial hypotheses to test with buyers.

The illustrative pilot totals BDT 517,500

A 12-week planning model with six exercise phones and 15% contingency.



Illustrative BDT assumptions, not market quotations. The report contains quantities, rates and exclusions.

Software evidence supports a physical pilot

The next gate is a supervised test on real phones.

Recorded evidence	Next acceptance evidence
60 journeys on each of two Android emulators	Three-phone relay and recovery
Signed receipt acceptance and rejection	Camera verification in airplane mode
Bundled bilingual and model workflows	Human task completion and accessibility
Local observer replay after disconnection	Phone continuity with the laptop off
Reproducible local verification scripts	Target-phone battery, memory and latency

Existing dated evidence. The 10,000-stream laptop test measures connection capacity, not physical mesh readiness.

The next step is one controlled exercise

A sponsor, an operational owner and a measured delivery scenario.

One referral chain.
A clear acceptance decision.

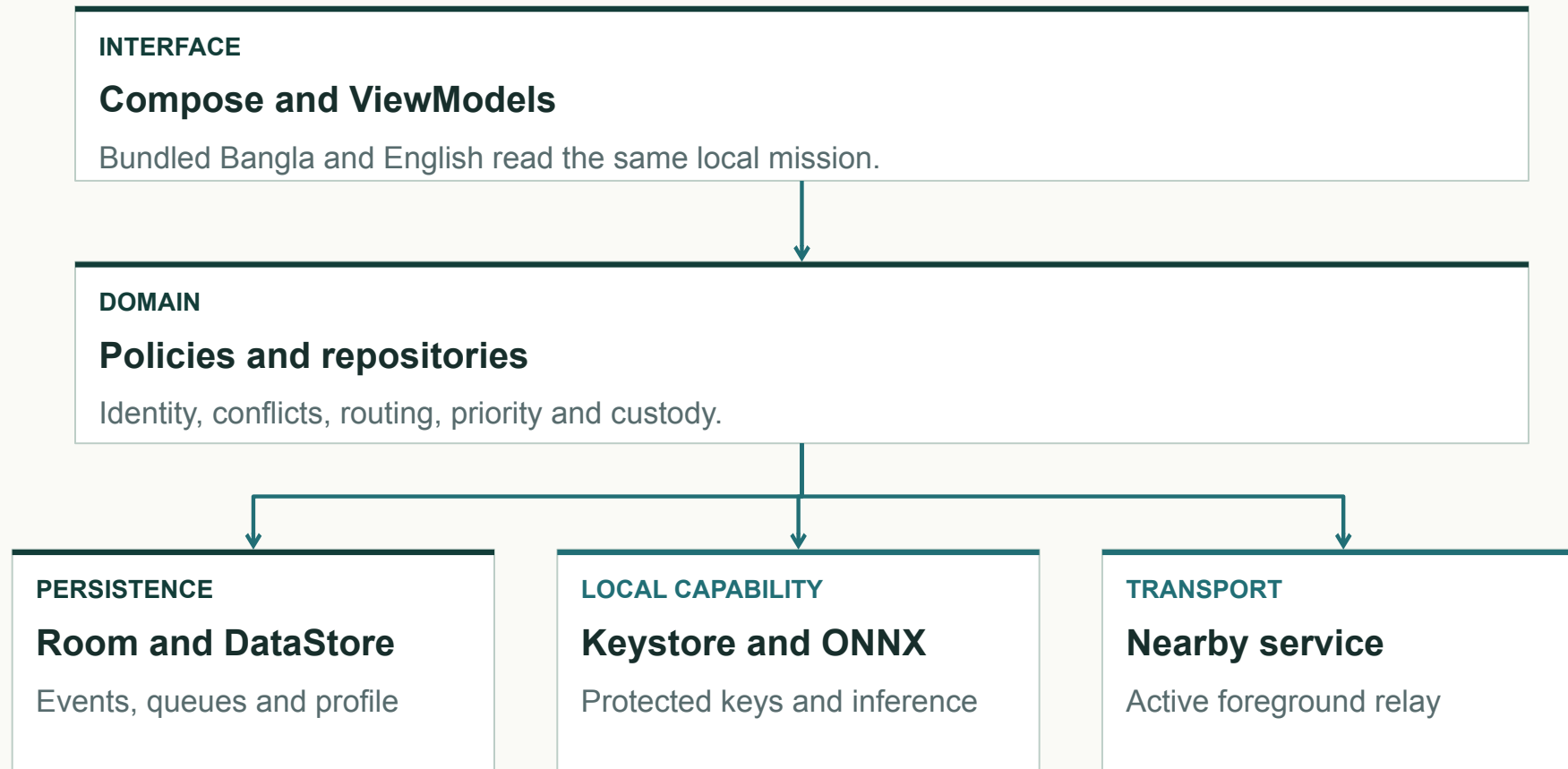
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Concept illustration

The phone owns its operational stack

Technical appendix

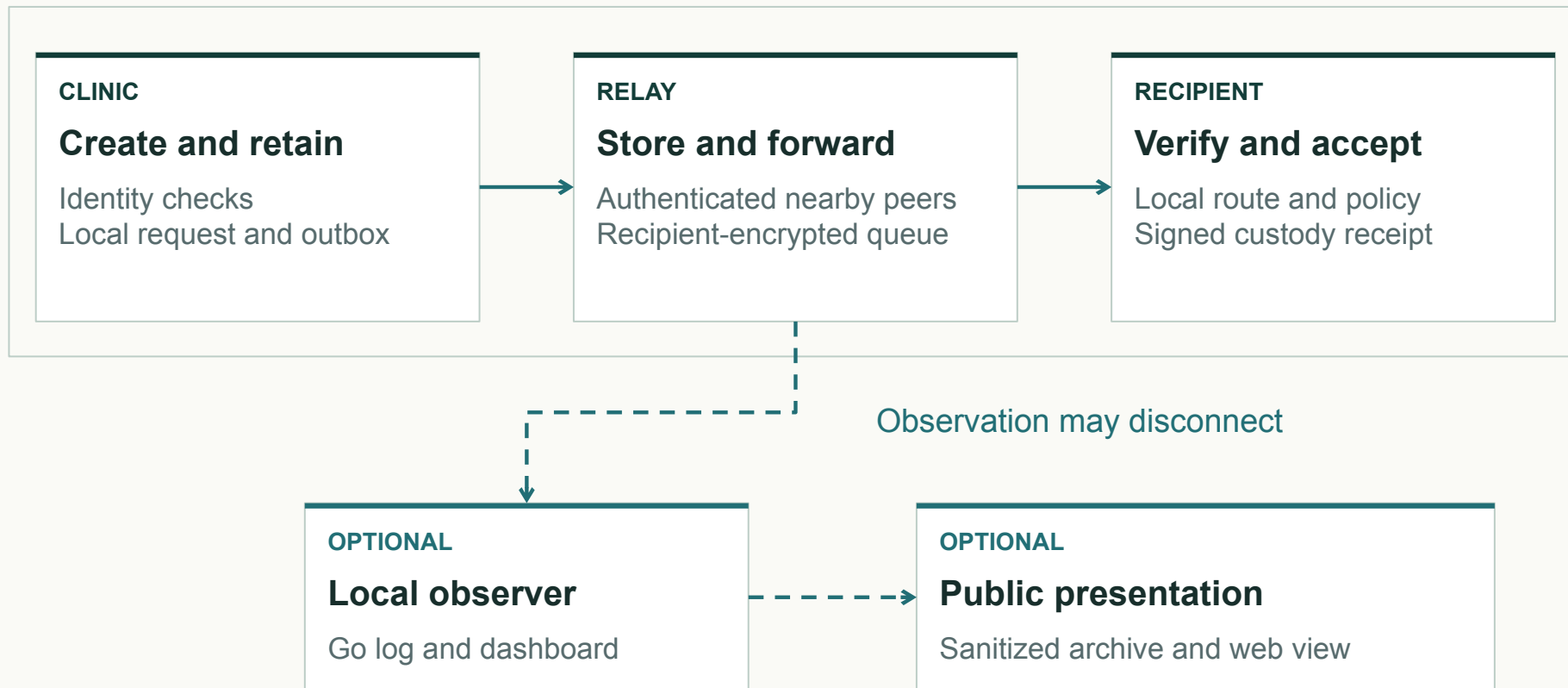


Original editable diagram derived from repository architecture and documented acceptance boundaries.

Optional observation stays outside field authority

Technical appendix

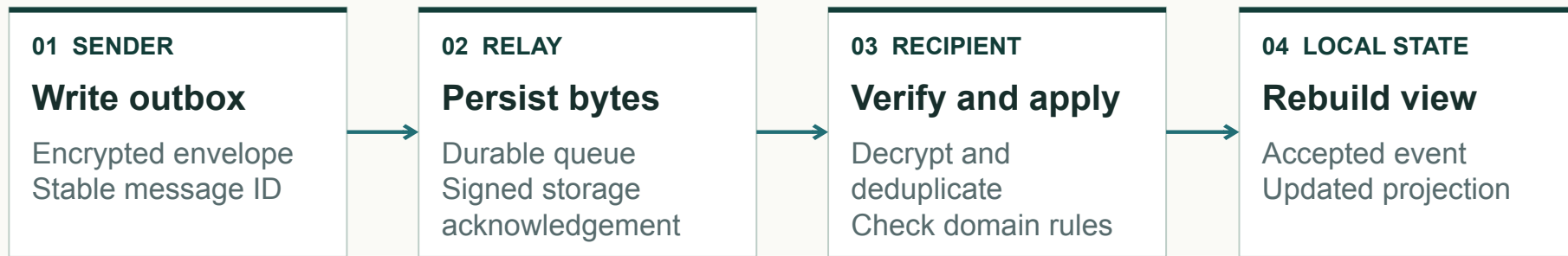
FIELD AUTHORITY / Bundled Bangla and English



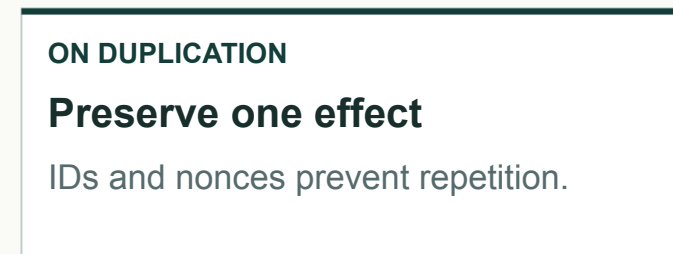
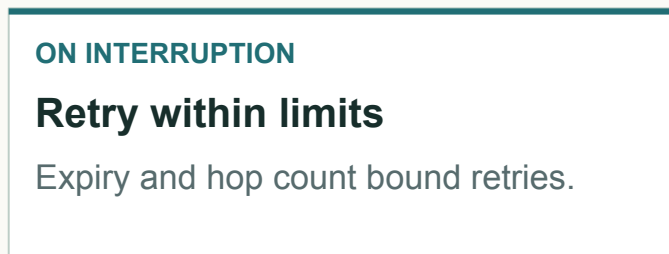
Original editable diagram derived from repository architecture and documented acceptance boundaries.

Acknowledgement precedes domain acceptance

Technical appendix

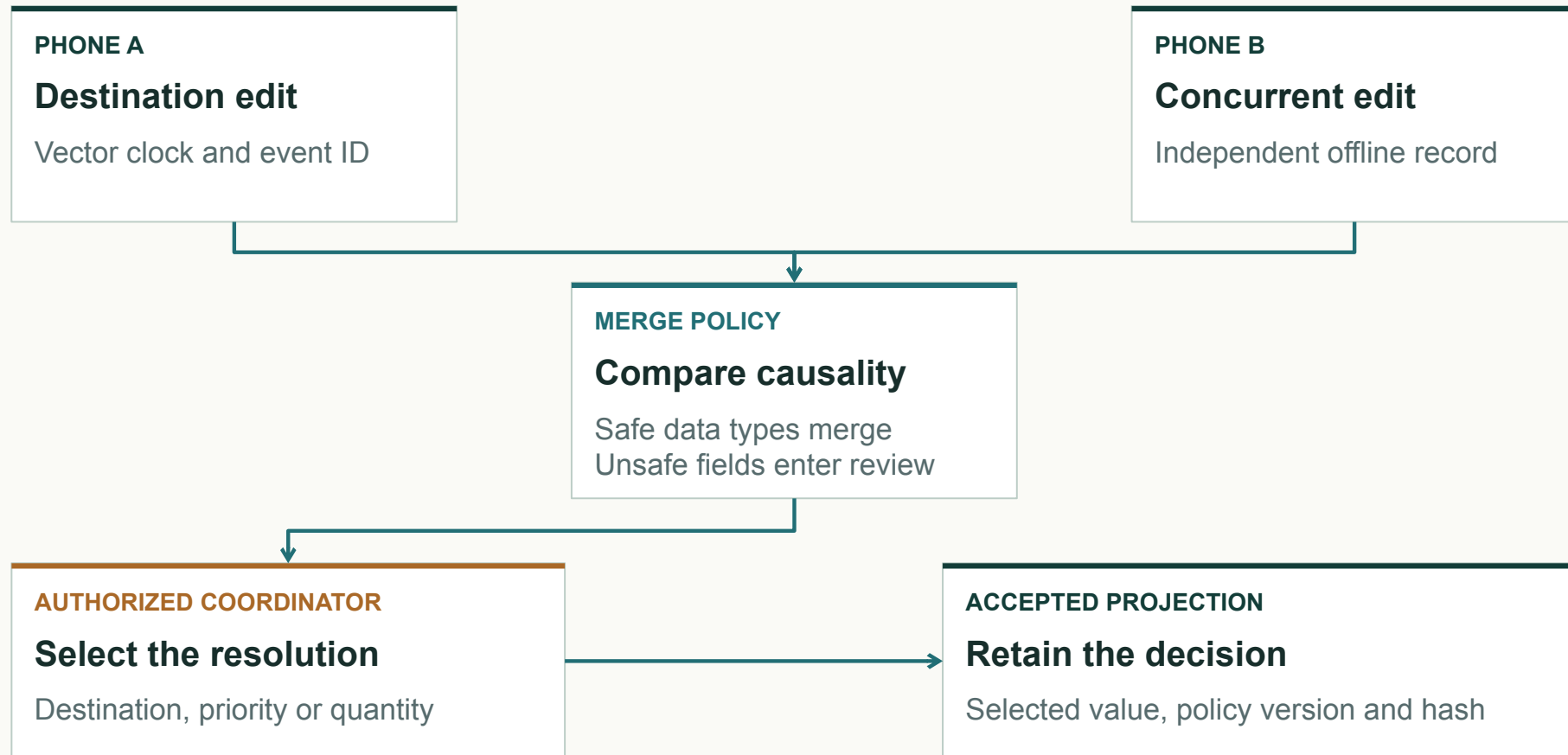


A stored envelope may still fail recipient or domain verification.



Unsafe concurrent edits need an authorized decision

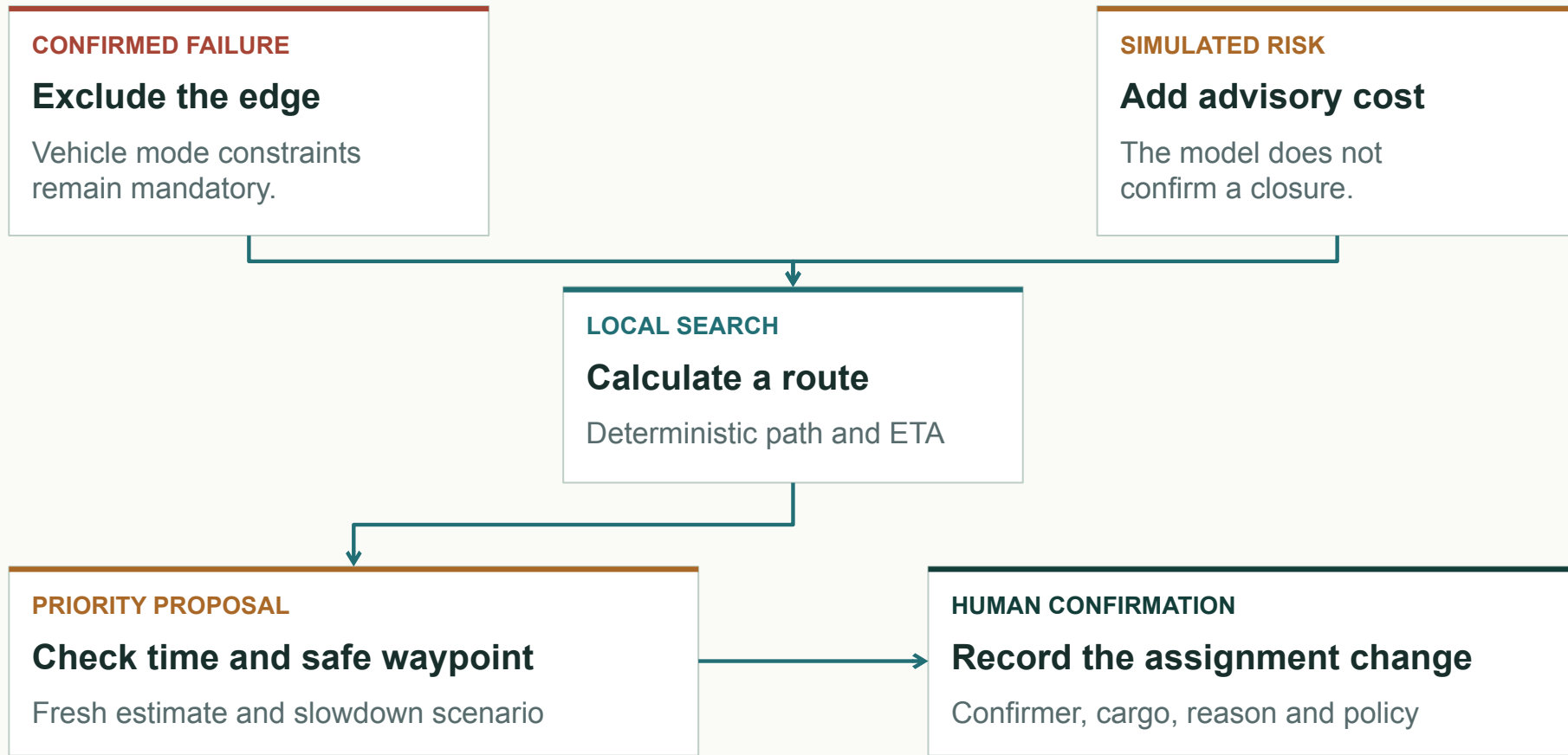
Technical appendix



Original editable diagram derived from repository architecture and documented acceptance boundaries.

Route failures and risk have different effects

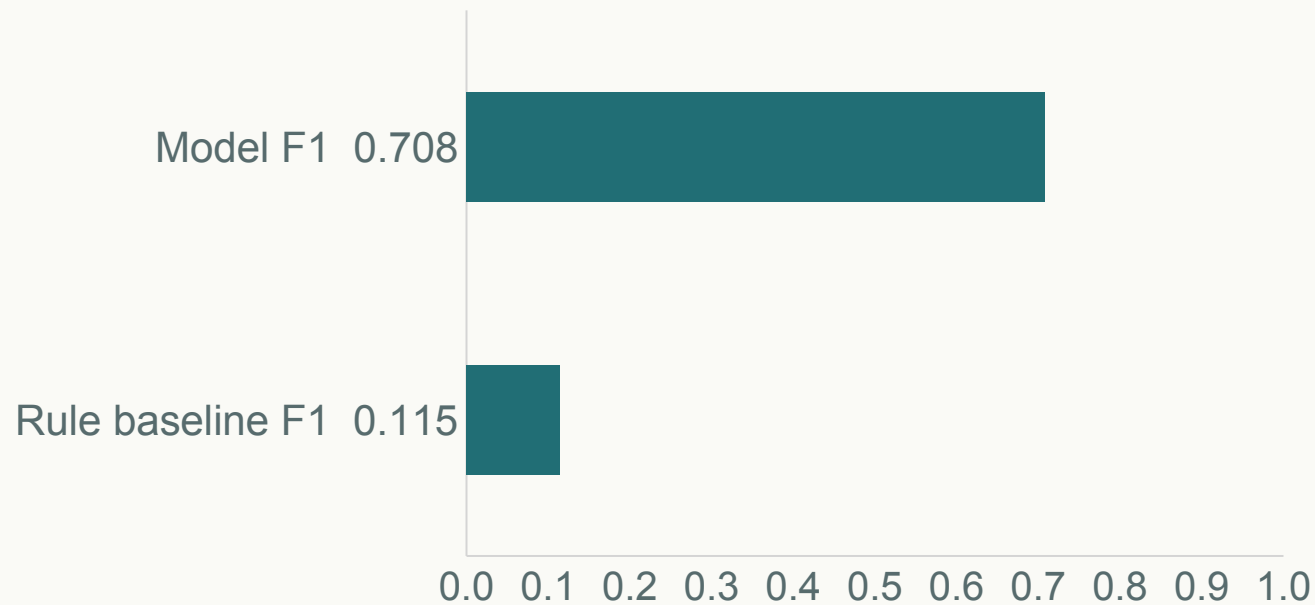
Technical appendix



Original editable diagram derived from repository architecture and documented acceptance boundaries.

The model improves recall on synthetic labels

The result demonstrates integration and an explicit error tradeoff.



Recall 0.837

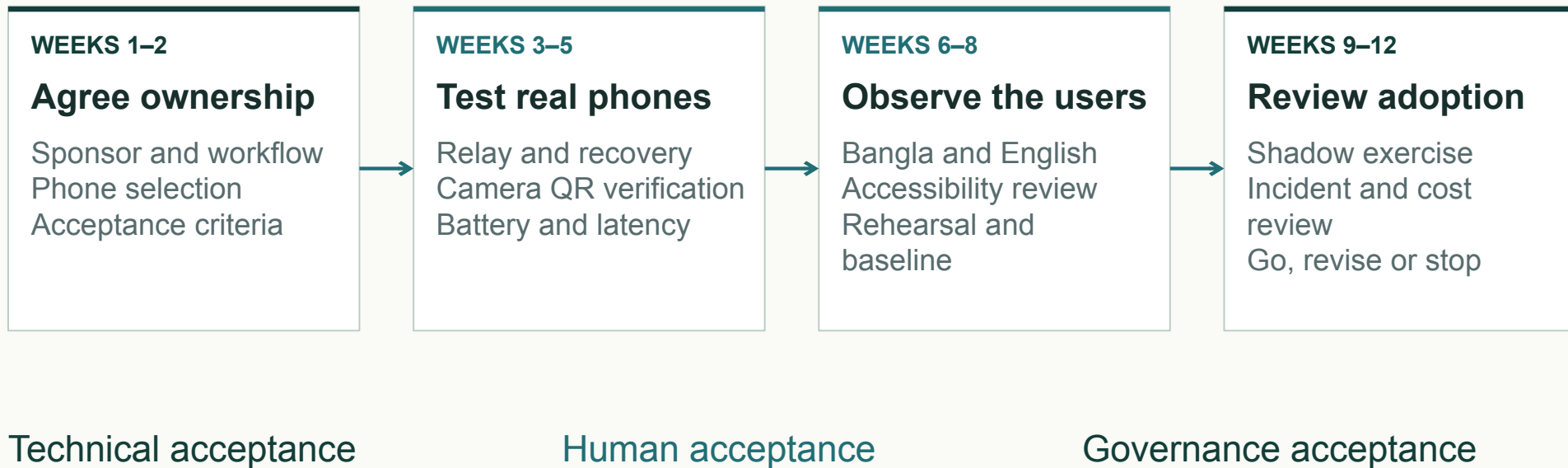
The model finds more synthetic positives, with more false alarms.

900 held-out synthetic examples

Actual model-card metrics. Real observations, calibration and geographic/time holdouts remain future research.

Pilot milestones support an adoption decision

Proposed twelve-week exercise



Proposed schedule. No customer commitment or organizer requirement is implied.

Technical, human and governance acceptance determine whether expansion is justified.

Market sizing starts with verified institutions

Geographic need provides context rather than a revenue total.



Bottom-up sizing follows verified organizations, budgets and purchase cycles.

S03: NAWG June 2024 assessment. Revenue market size is not yet established.

Discovery tests the buying hypothesis

Twelve proposed interviews across operational and institutional roles.

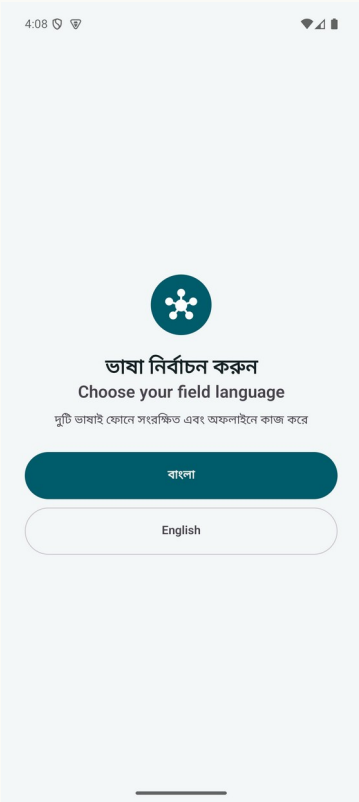
Participants	What to learn	Decision it informs
4 field users	Last outage, current workaround, critical errors	Workflow and interface fit
4 coordinators	Decision authority, custody and handoff exceptions	Operating procedures
4 budget or IT owners	Budget timing, procurement, support and data rules	Buyer and service model

Recruit across Bangla and English use, device experience and accessibility needs.

Illustrative discovery design. Demand, willingness to pay and sponsor ownership remain unvalidated.

Both languages share the same mission

Product appendix: language choice and authority.



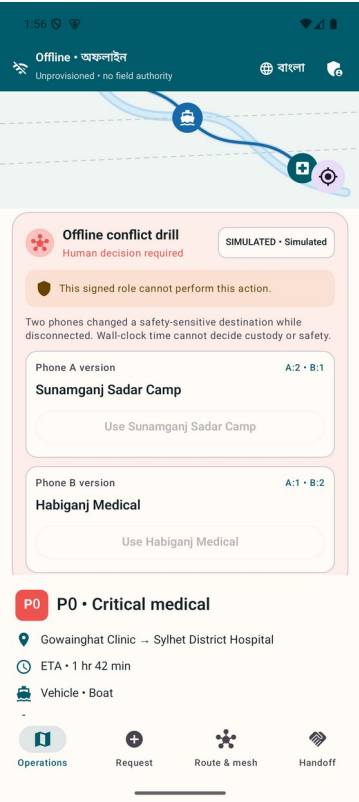
Language choice



Bangla home



Trusted identity

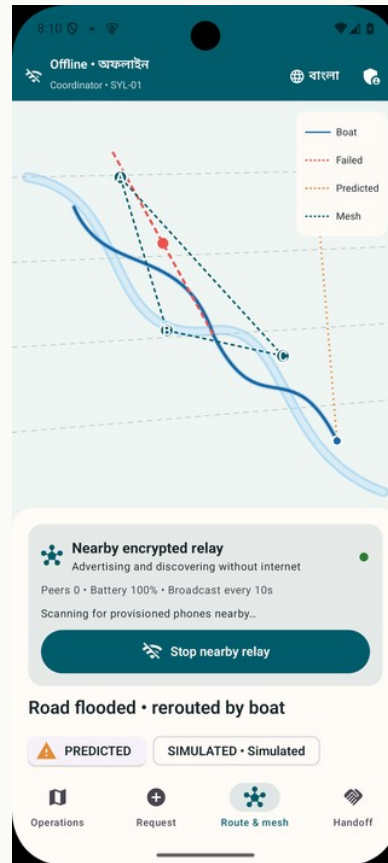
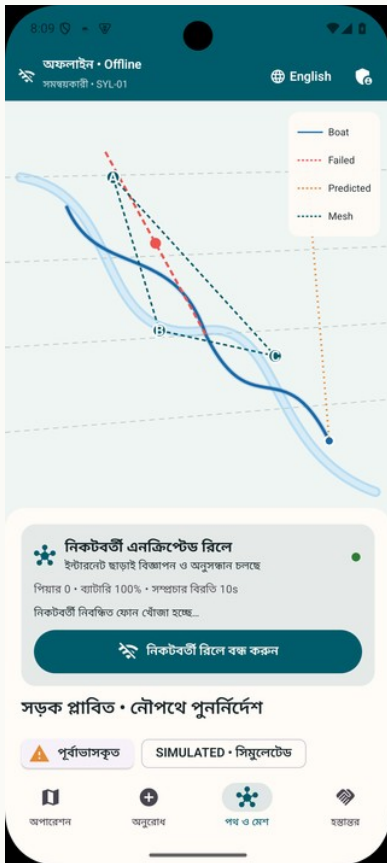


Role denial

Original full captures. Language parity alone does not establish human accessibility.

The active relay has a visible lifecycle

Product appendix: Bangla and English Nearby sessions.



Authenticated peers

Nearby connection acceptance is followed by independent signed challenge verification.

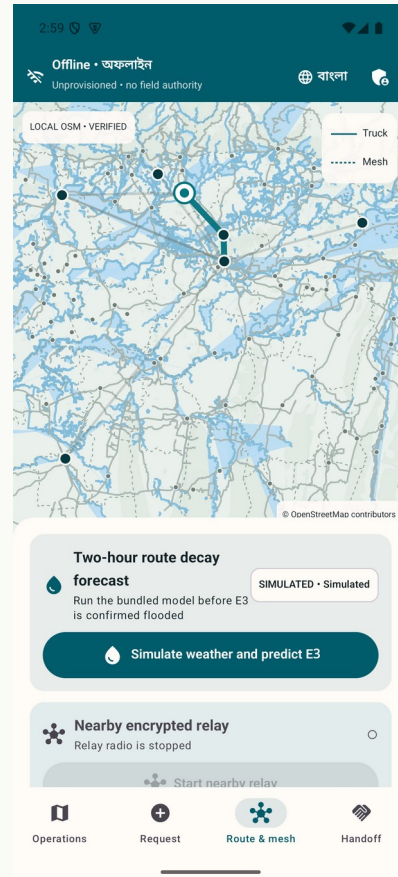
Durable transfer

Queues persist before acknowledgement. Interruption and retry require physical validation.

Full source captures. Three physical phones must demonstrate relay interruption and recovery.

Maps remain available on the phone

Product appendix: geographic provenance and simulated overlays.



Local geography

Reviewed extract and checksum verification support offline rendering.

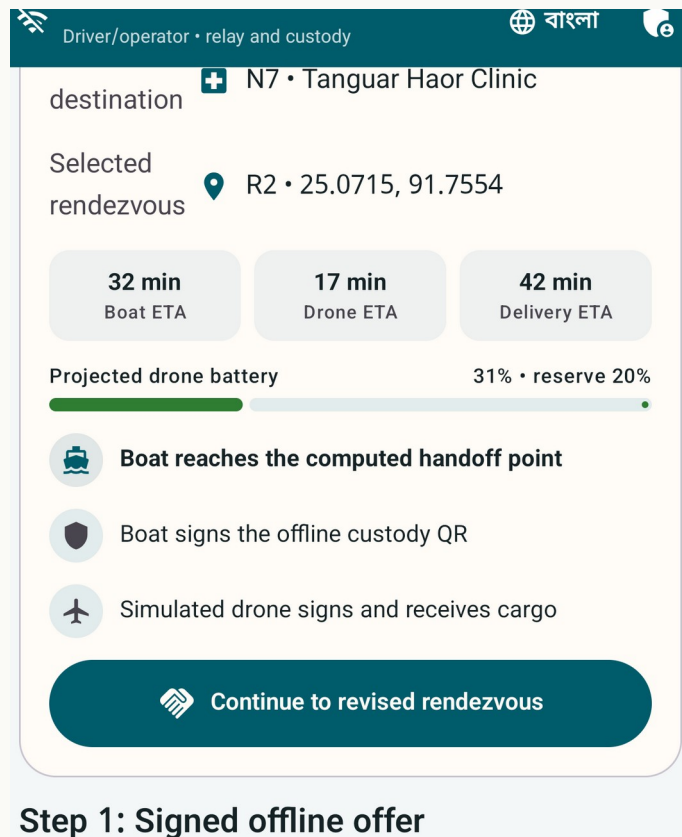
Separate sources

OSM attribution identifies the basemap. The mission overlay remains simulated.

OpenStreetMap contributors. Full source captures. Deployment requires maintained maps and closure procedures.

A simulated delay moves the rendezvous

Product appendix: the hybrid fleet calculation.



18-minute delay

The seeded calculation moves the meeting point from R3 to R2.

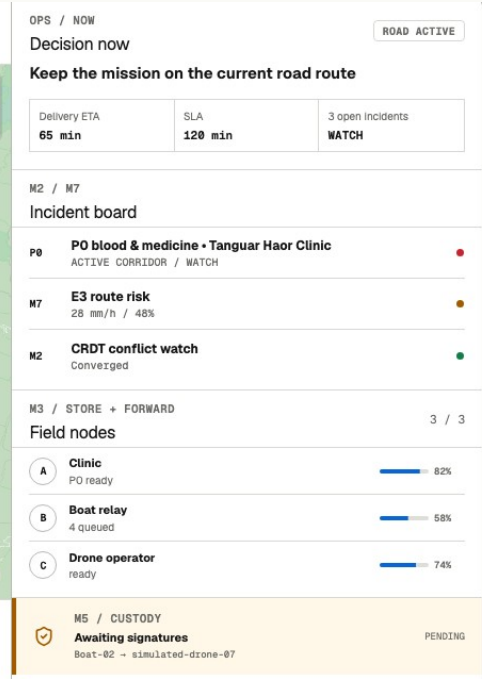
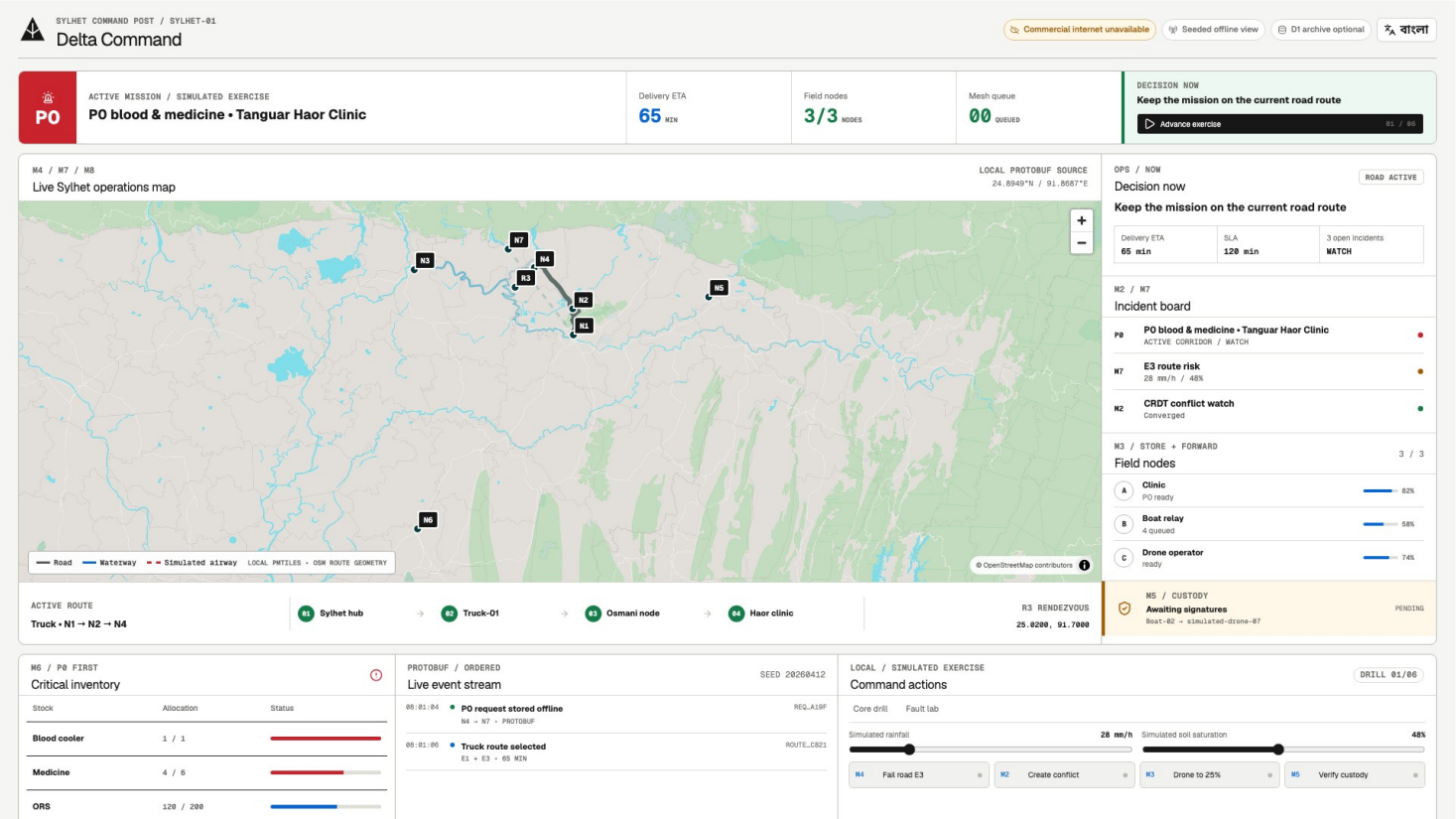
20% battery reserve

Candidate selection retains the configured reserve before handoff.

Enlarged source crop. All vehicles and telemetry are simulated. No drone hardware or control is included.

The observer makes the mission readable

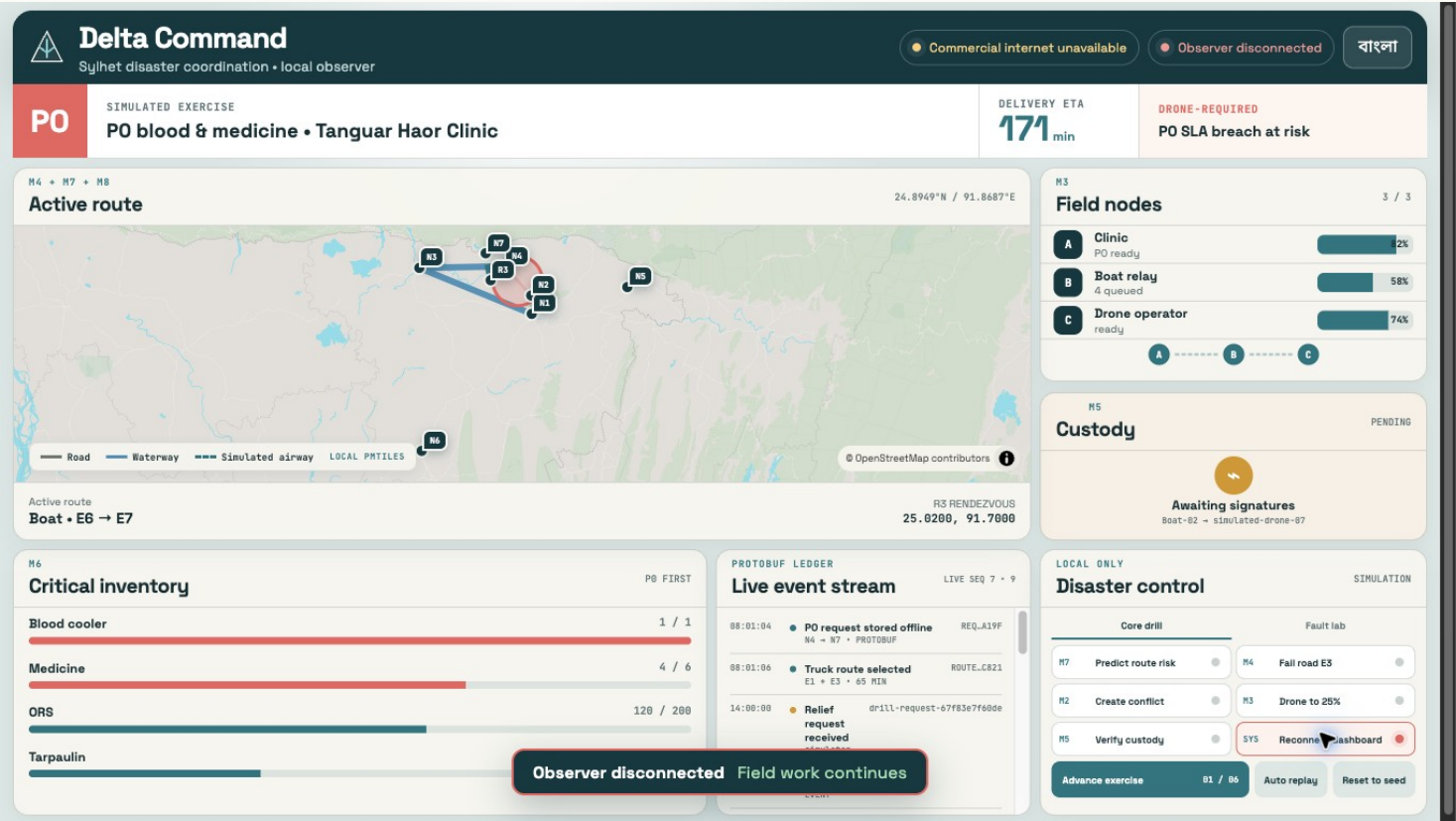
Product appendix: overview and enlarged decision detail.



Full capture plus enlarged detail. OpenStreetMap contributors. Field actions do not depend on this screen.

The observer can replay a missed interval

Product appendix: disconnect at sequence 7, then recover events 8–14.



Sequence 7

Observer disconnects

Events 8–14

Replay after reconnect

Actual local observer evidence. This does not establish the physical laptop-off phone journey.

Sources and evidence

The report contains the complete register and implementation boundaries.

Problem and market

UN Bangladesh 2022 response plan
NAWG June 2024 assessment
World Bank May 2025 financing

Implementation

Architecture and testing documents
Dated local gate and model card

Comparison

Sahana, KoboToolbox, Ushahidi,
Briar and Meshtastic official sources

Traceability

Slide notes include sources.
Report pages 39–40 contain the register.